

Supplementary material for Statistical-mechanical characterization of memory and quench response in state-separated LLM-agent networks

S1. Finite-window persistence and order-parameter shape

The diagnostics below accompany equations (8)–(9) of the main text. Each quantity is computed within one complete model–graph–environment trajectory before uncertainty is summarized across the six clusters per model. The autocorrelation sum uses the fixed two-sweep truncation, while one- and three-sweep truncations remain machine-readable sensitivity analyses. The Binder statistic describes the shape of a finite $N = 16$ magnetization distribution. The source-data table additionally retains full-phase versus early/late half-phase estimates and cluster-mean versus pooled-moment rules. It is not a crossing analysis or evidence of a phase transition.

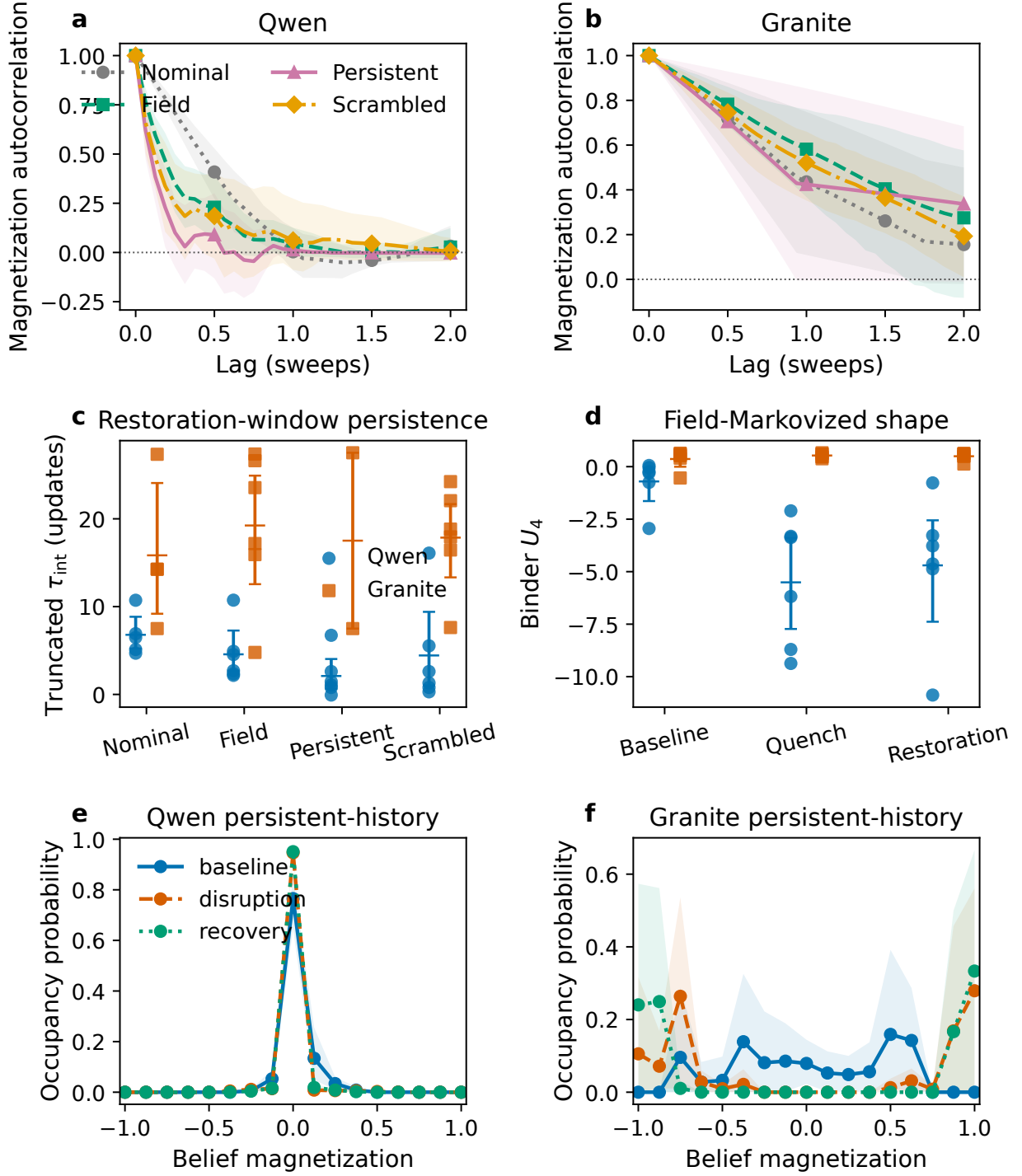


Figure 1: Finite-window persistence and order-parameter shape. Autocorrelation curves and their 95% intervals compare the nominal late window with the three field arms using complete clusters; integrated correlation sums show all cluster values at the fixed two-sweep truncation. Binder cumulants are paired with empirical belief-magnetization occupancies so their finite-sample meaning remains visible. These diagnostics do not establish critical slowing down or a phase transition.